

Gap Analysis — Supporting Material

Prepared from the Existing Solutions review. Intended as input to the Gap Analysis section of the report.

1. How This Was Derived

Six existing systems were reviewed against eight criteria drawn from what Itqan is designed to do, with every finding taken from the vendor's own website, help centre, app-store listing or privacy policy. This document reports what that review found to be **already available** and what it found to be **absent from the public record**.

Two rules govern what follows, and they matter because the conclusions run in both directions:

- A capability is treated as present if a vendor documents it, whether or not Itqan also intends to provide it.
- A capability is treated as absent only where the public record contains nothing. Absence of published evidence is reported as such, not as proof that no system has it.

2. What Is Not a Gap

This section exists to prevent the report claiming novelty it cannot defend. Each item below was initially assumed to distinguish Itqan and was found not to.

Assumed differentiator	What the review found
Real-time word-level error detection	Tarteel has provided this since 2018, advertising detection of "missed, incorrect and skipped words in your recitation"
Showing the error on the word in the text	Tarteel highlights the word and lets the user tap it to see "exactly what you recited versus the correct Ayah"
Playing the correct recitation of a wrong word	Tarteel: "Tap the reciter button next to it to hear the correct recitation." Tilawa provides word-by-word reference audio
Smart Prompting — helping a reciter who pauses	Tarteel states "if you get stuck, our A.I will be sure to let you know," and its Testing mode provides reveal controls in which "each reveal counts as a mistake." This is more developed than Itqan's concept currently describes
Tajweed feedback as a capability	Marketed by Tilawa, Thurayya and TajweedMate, and returned by the Qurani.ai API as a separate field from letter errors
Teacher and class monitoring	Tasmi allows teachers and parents to "assign passages and confirm practice happened"
A Juz Amma scope	Thurayya is scoped to Juz Amma and directed at children
Streaks, progress tracking, session scores	Present across most reviewed systems; Tarteel tracks memorization "all the way to the individual word level"

Consequence: Itqan is not the first or the only AI-based Quran recitation system, and none of the above can be presented as a novel contribution of this project.

3. The Gaps That Survive

Four findings held up when the full set of reviewed systems was considered.

Gap 1 — No system publishes an accuracy figure for recitation error detection

The only public accuracy figure found for any reviewed system is a **4% word-error rate for Arabic transcription**, quoted by Tarteel's chief executive in an NVIDIA case study. Transcription word-error rate measures how accurately speech is converted to text. It does not measure how reliably errors in recitation are detected, and no vendor connects the two.

Tarteel's own help centre concedes both false positives and a specific weakness, stating that *"Madd Lazim for example is something our model does not recognize well."* Tilawa's terms state only that it does *"not guarantee that the AI analysis will be 100% accurate at all times."*

What Itqan can claim: that it will report a measured accuracy figure with a stated test set and results broken down by error type. The bounded scope of 37 surahs is what makes such an evaluation feasible within a graduation project.

What must not be claimed: that competitors are inaccurate. No published figure is not evidence of poor performance, and the transcription figure must never be presented as a detection figure.

Gap 2 — No system publishes a latency figure

No figure for how quickly feedback arrives appears on any vendor page reviewed. Both Tarteel and Tilawa process audio on their own servers; Tilawa states *"the audio is sent to our secure servers for AI-powered analysis."*

What Itqan can claim: that it will measure and report end-to-end feedback latency under stated conditions.

What must not be claimed: that Itqan is faster than any existing system. No baseline has been published and no comparative measurement has been made, so such a claim could not be supported or tested.

Gap 3 — Tajweed rule sets are not published alongside evidence of reliability

Tarteel states plainly that it does not attempt tajweed-rule detection: *"Our AI is currently not trained to recognize tajweed mistakes yet unfortunately."* Tilawa, Thurayya and TajweedMate all advertise it — Tilawa claiming *"over 16 tajweed rules"* — but none publishes which rules are covered, how they are detected, or how reliably.

No reviewed system publishes both its rule set and an evaluation of that rule set.

What Itqan can claim: that it implements tajweed correction over a documented, bounded rule list for Juz Amma and reports how each rule was evaluated. The contribution is the stated rule set together with its evaluation, not the existence of tajweed feedback.

What must not be claimed: that Itqan is the first or only system to correct tajweed. Three of the six reviewed systems market exactly that.

Gap 4 — Control over one's own recitation audio is not free

Tarteel's opt-out from having recordings used for AI training is *"available for Premium users only,"* deletion of stored recordings is likewise a paid feature, and its privacy policy — last updated May 2023 —

states no retention period for recitation audio.

What Itqan can claim: that consent, a stated retention position, deletion and any training opt-out are available to every user at no cost. Itqan goes further by not retaining recitation audio at all.

What must not be claimed: that competitors are careless with privacy. Both publish privacy policies, and Tilawa's stated position that audio is *"not permanently stored after analysis"* is a strong one.

4. The Defensible Position

The gaps above share a shape, and it is worth stating directly because it is what the Gap Analysis section is ultimately arguing.

Itqan does not introduce a capability that no product has. What the public record does not contain, across the systems reviewed, is **evidence about how well any of them work**: no published accuracy figure for error detection, no published latency figure, and no published tajweed rule set with a matching evaluation.

Itqan's position is therefore accountability rather than novelty. Its scope — 37 surahs, 564 verses — is small enough that accuracy and latency can actually be measured and reported rather than asserted, which is not a claim a full-Quran commercial product has any incentive to make. Alongside that, tajweed and pronunciation errors are typed and anchored to the exact word, a teacher role exists in the same system as the learner, and control over recitation audio is available to every user at no cost.

5. Wording Cautions

Four phrasings would not survive an examiner opening a competitor's website.

Do not write	Write instead
"Itqan is the first AI Quran recitation application"	"Itqan is one of several AI-based recitation systems; what distinguishes it is that its accuracy is measured and reported"
"Existing applications do not provide real-time feedback"	"Existing applications provide real-time feedback; none publishes how accurate it is"
"No system offers teacher monitoring"	"Teacher monitoring exists, for example in Tasmi; Itqan integrates it with AI-based error detection in one system"
"Itqan achieves high accuracy in tajweed detection"	"Itqan reports measured accuracy per tajweed rule, for the rules within its stated scope"

6. A Note on the Earlier Draft

An earlier version of this analysis examined only two systems, Tarteel and Tilawa, and on that basis claimed three gaps that do not survive: that no system documents a teacher-facing role, that no dedicated child-oriented mode exists, and that no system performs analysis offline. Reviewing the wider set contradicted all three — Tasmi provides teacher and parent monitoring, Thurayya is a dedicated children's product scoped to Juz Amma, and Hifz AI advertises offline operation.

This is recorded because it illustrates the risk the Gap Analysis section carries: a gap identified from a narrow sample is not a gap. The four gaps in Section 3 were re-checked against all six reviewed systems.